llmXive Automated Review of Co-Evolving Policy Distillation

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is **advisory, automated feedback for the authors**: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision

The review focuses on the accuracy of factual claims and their alignment with cited evidence.

The manuscript presents a coherent argument for CoPD, and most claims are well-supported by the provided data. However, there are minor issues in the phrasing of results in Sections 3.1, 3.2, and 2.3 that could lead to misinterpretation.

In Section 3.1, the sentence stating ”the Text-Expert's strong text capability (57.89) is only partially transferred, dropping to 56.09 in the distilled model” is structurally ambiguous. It grammatically suggests the Text-Expert's score dropped, which is impossible as it is a fixed teacher. Table 1 confirms 56.09 is the score of the *student* (Image branch) in the T->V direction. The text must be clarified to state that the *student's* capability dropped to 56.09.

In Section 3.2, the claim that ”Mixed RLVR achieves the highest video average among baselines” is technically imprecise. While Mixed RLVR (59.62) outperforms MOPD (58.32), it is compared against the Video-Expert (58.75) in the same table. The phrasing implies Mixed RLVR is the best overall baseline, which is true, but the distinction between ”consolidation baselines” and ”single-expert baselines” is blurred. Clarifying this scope would prevent confusion.

Finally, in Section 2.3, the authors state that as overlap approaches 1, ”OPD gain necessarily collapses to zero.” While this is a sound theoretical boundary condition, the pilot study only covers a limited range of overlap values and does not empirically verify this collapse. Presenting this as a confirmed fact rather than a theoretical expectation overstates the evidence provided in the pilot study. The language should be softened to reflect that this is a predicted outcome based on the KL divergence formulation.

These are minor writing and precision issues that do not invalidate the core science but require correction for factual clarity.

- [**writing**] In Section 3.1, the phrase 'Text-Expert's strong text capability... dropping to 56.09' is ambiguous. It implies the expert degraded, but Table 1 shows the student (Image branch) scored 56.09. Rephrase to clarify the student's score dropped.
- [**writing**] In Section 3.2, 'Mixed RLVR achieves the highest video average among baselines' is misleading as Video-Expert (58.75) is also a baseline. Specify 'among consolidation baselines' to distinguish from single-expert baselines.
- [**writing**] In Section 2.3, the claim that OPD gain 'necessarily collapses to zero' at overlap 1 is theoretical, not empirically shown in the pilot data. Soften to 'is expected to' or 'suggests' to reflect it is a hypothesis.

paper_reviewer_figure_critic

Verdict: major_revision_science
Figure 1
Figure 1 presents clear visualizations of training dynamics, but the caption contains grammatical errors and makes claims about monotonicity and stability that are contradicted by the plotted data lines.
Figure 2
Figure 2 provides a clear, high-level schematic of the CoPD method, effectively illustrating the workflow from initial model branching to the final merged model. The detailed insets for 'RLVR Update', 'OPD Update', and 'Mutual OPD' successfully clarify the specific mechanisms without clutter, and the diagram is self-contained with no missing labels or legends.
Figure 3
The figure contains four subplots, but the caption only describes three, completely omitting subplot (d) which displays the 'Symmetric KL' metric. Additionally, the caption conflates the data shown in subplots (c) and (d) into a single description for (c), failing to distinguish between the two separate metrics plotted against training steps.
Figure 4
Figure 4 combines cartoons and a bar chart to motivate CoPD, but the caption fails to reference the quantitative chart (d) and uses inconsistent terminology (GRPO vs RLVR) between the image labels and the text. The cartoon panels illustrate scenarios but do not explicitly visualize the 'limitations' claimed in the caption.

- [**science**] Figure 1: The caption for (a) claims the baseline overlap drops monotonically, but the 'Baseline · Text' line (orange open circles) clearly fluctuates and is not monotonic.
- [**science**] Figure 1: The caption for (b) states KL 'stays consistently low in [CoPD]', but the 'CoPD · Image' line (green solid squares) rises from ~0.02 to ~0.06, which is a 3x increase, not 'consistently low' compared to the baseline's 10x rise.
- [**writing**] Figure 1: The caption for (b) contains a grammatical error and missing subject: 'stays consistently low in .' (missing 'CoPD').
- [**science**] Figure 1: Panel (c) labels the x-axis 'CoPD S_RL : S_OPD' but includes a 'static

OPD' group which is not a ratio, creating a category error in the axis labeling.

- [**fatal**] Figure 3: The rendered image contains four subplots labeled (a), (b), (c), and (d), but the caption only describes three subplots (a), (b), and (c). Subplot (d) is completely missing from the description.
- [**fatal**] Figure 3: The caption describes subplot (a) as 'WeMath score before and after OPD' and subplot (b) as 'post-OPD gain plotted against top-k overlap', but the rendered image labels the top-left bar chart as (a) and the bottom-left scatter plot as (b). The caption text for (a) matches the top-left chart, but the caption text for (b) matches the bottom-left chart. However, the caption text for (c) describes 'top-k overlap drops and symmetric KL rises', which corresponds to the two right-hand plots i
- [**science**] Figure 3: The caption states that subplot (c) shows 'top-k overlap drops and symmetric KL rises', implying a single plot or a combined view. However, the image shows two distinct plots: (c) 'top-k overlap' vs 'Training step' and (d) 'Symmetric KL' vs 'Training step'. The caption conflates these two separate visualizations into one description for (c), leaving (d) undefined.
- [**writing**] Figure 3: The caption mentions 'across image and text branches' for subplot (c), which is consistent with the legend in the rendered plots (c) and (d). However, the caption does not explicitly mention subplot (d), creating a disconnect between the text and the visual layout which clearly separates the overlap and KL metrics.
- [**science**] Figure 4: The caption claims the figure illustrates 'limitations' of mixed-data RLVR (a) and static OPD (b), but the panels (a) and (b) only depict qualitative scenarios (student confusion/struggle) without quantitative data or explicit visual indicators of the specific limitations (e.g., performance drop, drift) mentioned in the text.
- [**writing**] Figure 4: The labels '(a) GRPO', '(b) OPD', and '(c) CoPD (Ours)' are placed below the cartoon panels, but the caption refers to '(a) mixed-data RLVR' and '(b) static OPD'. The term 'GRPO' in the image is not defined in the caption, creating a disconnect between the visual label and the textual description.
- [**writing**] Figure 4: The bottom bar chart (d) is not referenced in the caption, yet it contains the primary quantitative evidence ('achieving the best overall performance') supporting the caption's claim. The caption should explicitly reference panel (d) or the chart.

paper_reviewer_jargon_police

Verdict: minor_revision
The manuscript relies heavily on domain-specific acronyms and coined phrases that, while standard within the immediate RLVR sub-community, create barriers for the broader machine learning audience.

First, the Abstract introduces "RLVR" and "OPD" without expansion. While these are the core mechanisms, the abstract should define them as "Reinforcement Learning with Verifiable

Rewards" and "On-Policy Distillation" respectively upon first use to ensure accessibility.

Second, the paper frequently uses the phrase "behavioral distance" and "behavioral patterns" (e.g., Section 1, Section 3). While intuitive, these are vague jargon terms. The authors should consistently map these to the specific mathematical metrics they actually measure, such as "top-k token overlap" or "symmetric KL divergence," to avoid ambiguity. For instance, "closing the behavioral distance" in Section 1 should be phrased as "reducing the symmetric KL divergence between the teacher and student policies."

Third, the term "capability divergence" is used as a proper noun for a specific failure mode. It is not a standard term in general optimization literature. The authors should provide a concise, formal definition early in the Introduction to distinguish it from general gradient conflict or negative transfer.

Finally, the "hub-and-spoke topology" description in Section 2 is a metaphor that could be replaced with a more direct description of the data flow (e.g., "a central text branch exchanges distillation signals with peripheral image and video branches") to reduce cognitive load for non-specialists.

- **[writing]** The manuscript relies heavily on domain-specific acronyms and coined phrases that, while standard within the immediate RLVR sub-community, create barriers for the broader machine learning audience. First, the Abstract introduces "RLVR" and "OPD" without expansion. While these are the core mechanisms, the abstract should define them as "Reinforcement Learning with Verifiable Rewards" and "On-Policy Distillation" respectively upon first use to ensure accessibility. Second, the paper frequently use

paper_reviewer_logical_consistency

Verdict: minor_revision

"The paper presents a logically consistent argument for the Co-Evolving Policy Distillation (CoPD) method, with a clear chain of reasoning from the identified problem (capability divergence in mixed RLVR and low absorption in static OPD) to the proposed solution (co-evolving branches with mutual distillation) and experimental validation. The pilot study effectively supports the hypothesis that teacher-student behavioral overlap is critical for distillation efficiency, and the main results consistently demonstrate CoPD's superiority over baselines.

However, there is a significant logical inconsistency in the description of the 3-branch training topology. In Section 3.3 (\"Alternating Training Procedure\"), the text states: \"To avoid full pairwise distillation, we adopt a hub-and-spoke topology, where one branch serves as a shared hub and exchanges mutual OPD with each spoke branch.\" This implies a specific, sparse communication pattern (e.g., Text <-> Image, Text <-> Video, but not Image <-> Video). Yet, Algorithm 1 (lines 12-15) explicitly implements a full pairwise loop: \"for each j != k\", which would include distillation between the Image and Video branches. This contradiction between the textual description of the method's efficiency/scalability (hub-and-spoke) and the actual algorithmic implementation (full pairwise) undermines the logical consistency of the method's design claims. The authors must clarify whether the hub-and-spoke topology is used in the experiments (and update the algorithm accordingly) or if full pairwise distillation was used (and update the text to reflect this, potentially revising claims about scalability).

Additionally, while the claim that CoPD \"surpasses domain-specific experts\" is supported by the aggregate averages in Tables 1 and 3, the text in Section 4.1 could be slightly more precise. It states that in the T->V direction, \"the Text-Expert's strong text capability (57.89) is only partially transferred, dropping to 56.09\". While numerically correct, the phrasing might imply that the *distilled model's* text capability is 56.09, which is true, but the comparison to the *Text-Expert* (57.89) is the key point. The logic holds, but the phrasing could be tightened to avoid any ambiguity about which model's capability is being discussed.

Overall, the core scientific logic is sound, but the discrepancy in the 3-branch topology description requires correction to ensure the method is accurately and consistently presented."

- [**science**] Section 3.3 claims a 'hub-and-spoke topology' for 3-branch CoPD to avoid full pairwise distillation, but Algorithm 1 implements full pairwise loops (j != k). This contradiction invalidates the scalability claim and requires alignment between text and code.
- [**writing**] Section 4.1 phrasing regarding the T->V distillation drop (57.89 to 56.09) is ambiguous. Clarify explicitly that the distilled model's score is lower than the teacher's to prevent misinterpretation of the transfer efficiency.

paper_reviewer_overreach

Verdict: minor_revision

The paper makes several strong claims regarding the superiority of Co-Evolving Policy Distillation (CoPD) over existing paradigms, particularly the assertion that it "surpasses domain-specific experts" and establishes a "novel training scaling paradigm." While the experimental results are compelling, the manuscript occasionally extrapolates beyond what the specific data supports.

First, the claim in the Abstract and Conclusion that CoPD "surpasses domain-specific experts" is slightly over-claimed. In the two-branch setting (Table 1), CoPD achieves a Text Average of 58.76 compared to the Text-Expert's 57.89, and an Image Average of 56.97 compared to the Image-Expert's 55.76. While CoPD wins on the *averages*, the phrasing suggests a universal dominance. A more precise claim would be that CoPD achieves a superior *aggregate* performance or breaks the trade-off curve, rather than simply "surpassing" experts in a blanket statement. The distinction is subtle but important for scientific rigor.

Second, the Conclusion posits that the "model parallel training pattern offered by CoPD may inspire a novel training scaling paradigm." This is a significant extrapolation. The current experiments are limited to a single model size (Qwen3-VL-4B) and three specific modalities (text, image, video). Without evidence of how this scales to larger model sizes (e.g., 70B+) or a wider variety of capabilities, labeling it a "scaling paradigm" is premature. The authors should temper this to suggest it is a "promising direction for scaling" or "a potential paradigm," acknowledging the need for further validation at larger scales.

Finally, the Abstract claims CoPD achieves "all-in-one integration" and "significantly outperforming... domain-specific experts." While the results show CoPD beats the *static* experts in the consolidated model, the paper acknowledges that Mixed RLVR actually achieved a higher Video Average (59.62) than CoPD (59.21) in the three-branch setting (Table 2), albeit at the cost of text performance. The ”all-in-one” claim risks obscuring the fact that a single-modality focus (Mixed RLVR on video) can still yield higher raw performance in that specific domain. The authors should clarify that CoPD is the optimal solution for *balanced* multi-capability consolidation, rather than implying it is the absolute peak for every individual capability in isolation.

- [**writing**] The claim that CoPD ’surpasses domain-specific experts’ (Abstract, Conclusion) is over-claimed. Table 1 shows CoPD beats Image-Expert but only slightly exceeds Text-Expert averages. The text implies a universal breakthrough where data shows a nuanced trade-off. Qualify this to ’surpasses experts in aggregate’ or ’specific benchmarks’.
- [**writing**] The conclusion states CoPD ’even outperforming the respective expert models’ and suggests a ’novel training scaling paradigm.’ This extrapolates beyond evidence from a single 4B model on three domains. The claim of a general ’scaling paradigm’ is premature without ablation on model size. Temper language to ’promising direction’ rather than a definitive new paradigm.
- [**writing**] The abstract claims ’all-in-one integration’ significantly outperforming experts. However, Table 2 shows Mixed RLVR achieved a higher Video Average (59.62) than CoPD (59.21), albeit at the cost of text. The ’all-in-one’ claim glosses over that Mixed RLVR was better at video alone. Clarify that CoPD is the best *balanced* solution, not the absolute peak for every modality in isolation.

paper_reviewer_safety_ethics

Verdict: minor_revision
The manuscript presents a novel method for consolidating multiple expert capabilities in large language models. From a safety and ethics perspective, the paper is generally sound but lacks necessary disclosures regarding data provenance and potential misuse.

First, while the authors utilize public benchmarks (AIME, HMMT, MMMU) and open-source datasets (Polaris, MMFineReason), the **Evaluation** section (Section 4.1) does not explicitly state compliance with the specific licenses of these datasets or confirm that the video data filtering process (using Qwen3-8B-VL) did not inadvertently retain personally identifiable information (PII) or sensitive content. Given the use of video data, which often contains human faces or private environments, a statement confirming that the dataset was screened for privacy violations or that it strictly adheres to the terms of the source repositories (e.g., OneThinker, VideoChat-R1) is necessary.

Second, the **Conclusion** and **Introduction** heavily emphasize the concept of ”Self-Taught RLVR” and models ”teaching themselves.” While technically accurate in the context of on-policy distillation, this language risks anthropomorphizing the system and obscuring the human effort involved in curating the training data. To maintain transparency, the authors should clarify that the ”self” is a proxy for the model’s internal state and that the knowledge base is entirely derived from human-generated or human-curated data, preventing the misconception that the model is autonomously generating new knowledge without human oversight.

Finally, the paper claims state-of-the-art performance on competition-level mathematics and reasoning benchmarks. There is no discussion of **dual-use risks**. A model with enhanced reasoning capabilities could potentially be used to automate the solving of standardized tests (undermining educational integrity) or to generate sophisticated adversarial prompts against other AI systems. The authors should include a brief ”Safety and Limitations” section acknowledging these potential misuse scenarios and suggesting that future work should include safety alignment or red-teaming of the co-evolved models.

These are primarily writing and disclosure issues that do not invalidate the scientific contribution but are essential for responsible publication.

- [**writing**] The paper lacks an explicit statement regarding Institutional Review Board (IRB) or ethics committee approval for the use of human-generated data (e.g., AIME, HMMT, MMMU) and the filtering of video datasets. While these are public benchmarks, the methodology section (Section 4.1) should explicitly confirm that the data usage complies with the original licenses and that no private or sensitive user data was inadvertently included in the filtered video sets.
- [**writing**] The 'Self-Taught RLVR' series claims models can 'self-evolve' and 'teach themselves' (Conclusion). This framing risks obscuring the human labor and curation involved in dataset creation (e.g., Polaris, MMFineReason). The authors should add a clarification in the Limitations or Ethics section acknowledging that the 'self' is trained on human-curated data and that the system does not possess autonomous agency, to prevent misinterpretation of the model's capabilities.
- [**writing**] The paper reports significant performance gains on high-stakes reasoning benchmarks (AIME, HMMT). There is no discussion of potential dual-use risks, such as the model being used to generate solutions for unauthorized exams or to automate the creation of adversarial examples against other reasoning systems. A brief 'Safety and Limitations' paragraph addressing these potential misuse scenarios is required.

paper_reviewer_scientific_evidence

Verdict: minor_revision

The scientific evidence supporting the central claims of Co-Evolving Policy Distillation (CoPD) is generally sound in its experimental design but lacks necessary statistical rigor to fully substantiate the magnitude of the reported improvements.

The pilot study in Section 3.3 (motivation-new.tex, lines 145-165) is critical for the paper's premise, asserting that OPD gain is linearly correlated with teacher-student behavioral overlap ($r=0.89$). While the trend is plausible, the evidence is currently anecdotal. The text mentions varying sampling temperatures to generate students but fails to specify the number of data points (N) used to calculate this correlation. Without the sample size and a reported p-value, it is impossible to determine if this correlation is statistically significant or an artifact of a small, cherry-picked set of checkpoints. This is a foundational claim; if the correlation is not robust, the motivation for ”co-evolving” rather than ”static” distillation weakens.

Furthermore, the main experimental results (eval.tex, Tables 1 and 2) present a potential issue regarding effect size and statistical significance. The paper claims CoPD ”significantly outperforms” baselines and even ”surpasses domain-specific experts.” However, the absolute gains over the strongest single-domain experts are often small (e.g., Text Avg: 58.76 vs. 57.89, a ~1.5% relative gain). In the context of large-scale RL training, such margins can easily fall within the variance of different random seeds or evaluation stochasticity. The manuscript does not report standard deviations, confidence intervals, or results from statistical significance tests (e.g., paired t-tests across benchmarks). Without this, the claim that CoPD definitively breaks the ”capability ceiling” is not fully supported by the provided evidence.

Finally, the ablation study (eval.tex, Table 3) attempts to isolate the contribution of the ”merge” operation. The authors claim that even without merging, individual branches outperform static OPD. While the numbers in Table 3 (e.g., Text-Branch Only: 57.24 vs. OPD_V->T: 56.09) support a directional improvement, the margin is narrow. The evidence would be stronger if the authors provided error bars or repeated the ablation with different random seeds to demonstrate that the ”co-evolution” effect is stable and not a fluke of a specific initialization.

To strengthen the scientific evidence, the authors should: (1) report the sample size and p-value for the pilot study correlation; (2) include statistical significance testing (e.g., confidence intervals) for the main benchmark results to validate the ”surpassing experts” claim; and (3) clarify the stability of the ablation results.

- [**science**] The pilot study in Section 3.3 (Fig 4) claims a strong linear correlation (r=0.89) between top-k overlap and OPD gain. The text describes varying student temperature to create this data, but does not specify the number of checkpoints sampled or the statistical significance (p-value) of the correlation. Please report the sample size (N) and p-value to validate the robustness of this motivational claim.
- [**science**] In the main results (Tables 1 & 2), CoPD is claimed to ’surpass domain-specific experts.’ However, the reported gains over the strongest single-expert baselines (e.g., Text-Expert on Text Avg) are often marginal (e.g., 58.76 vs 57.89). The paper lacks a statistical significance test (e.g., paired t-test or bootstrap confidence intervals) across the multiple benchmarks to confirm these improvements are not due to random variance in the evaluation.
- [**science**] The ablation study (Table 3) removes the ’merge’ operation and evaluates single branches. The text claims these branches ’already surpass’ static OPD. However, the table shows the ’Text-Branch Only’ (57.24) is only marginally better than OPD_V->T (56.09) and comparable to OPD_T->V (56.29). The evidence for the ’co-evolution alone’ claim is weak without error bars or a clearer statistical comparison against the static baselines.

paper_reviewer_statistical_analysis

Verdict: minor_revision

The statistical analysis presented in the manuscript is generally aligned with the methodological claims, particularly the pilot study motivating the behavioral consistency hypothesis. However, several critical statistical details are missing that prevent a full assessment of the robustness and significance of the reported results.

First, in Section 3.3 (Pilot Study), the authors report a strong linear correlation ($r=0.89$, $R^2=0.79$) between teacher-student top-k overlap and OPD gain (Figure 2b). While this supports the hypothesis, the manuscript fails to report the sample size (N) of the student checkpoints used to generate this data point. Without N, it is impossible to calculate the p-value or confidence interval for the correlation coefficient. A high R^2 on a small sample (e.g., $N < 10$) can be misleading and may not be statistically significant. The authors must explicitly state the number of data points and provide a significance test to validate this claim.

Second, the main results in Tables 1, 2, and 3, as well as the ablation study in Table 3, report point estimates for accuracy (e.g., 58.76 vs. 57.41). Large language model evaluation on benchmarks often exhibits variance due to stochastic decoding and prompt sensitivity. The manuscript does not report standard deviations, confidence intervals, or results of statistical significance tests (such as paired t-tests or bootstrap resampling) across multiple random seeds. Without these, it is unclear whether the observed improvements (some of which are marginal, e.g., $< 1\%$) are statistically distinguishable from noise or if they represent genuine methodological gains.

Finally, the implementation details mention sampling 8 rollouts per prompt but do not specify the number of independent evaluation runs (seeds) performed for the final benchmark scores. To ensure reproducibility and statistical rigor, the authors should report the mean and standard deviation of the accuracy across at least 3-5 independent runs for each configuration. This is standard practice in RLVR literature to distinguish signal from stochastic variance.

- [**science**] The pilot study in Section 3.3 reports a linear correlation (r=0.89, R^2=0.79) between top-k overlap and OPD gain. The manuscript must report the sample size (N) of the student checkpoints used to derive this correlation and provide a p-value or confidence interval to establish statistical significance, as a correlation alone does not confirm the relationship is not due to chance.
- [**science**] In the ablation study (Table 3), performance drops are reported (e.g., text reasoning dropping from 58.76 to 57.41). The paper lacks statistical significance testing (e.g., paired t-tests or bootstrap confidence intervals) to determine if these differences exceed the variance inherent in LLM evaluation, especially given the small magnitude of some improvements.
- [**science**] The experimental setup mentions sampling 8 rollouts per prompt at temperature 1.0. To ensure reproducibility and statistical robustness, the authors should specify the number of independent evaluation runs (seeds) performed for each benchmark and report the standard deviation of the mean accuracy, rather than just the mean.

paper_reviewer_writing_quality

Verdict: minor_revision

The manuscript demonstrates a high level of overall clarity and logical flow, effectively communicating the motivation, methodology, and results of the Co-Evolving Policy Distillation (CoPD) framework. The narrative structure is strong, particularly in the Introduction and Motivation sections, where the problem of capability divergence and the limitations of static distillation are clearly articulated. The transition from the pilot study to the proposed method is smooth and well-supported by the text.

However, there are a few minor writing issues that should be addressed to ensure professional polish. In Section 4.1 (Implementation Details), the word ”specifc” appears twice as a typo for ”specific.” Additionally, in Section 3 (Introduction), there is a stray comment marker (%) that interrupts the sentence flow regarding gradient conflicts. While these do not significantly impede understanding, correcting them will improve the manuscript’s readability. The abstract and conclusion are well-written and concise, effectively summarizing the contributions. The use of technical terms is consistent, and the mathematical notation is clearly defined. Overall, the writing quality is strong, with only minor corrections needed.

- **[writing]** In Section 4.1 (Implementation Details), the word ’specifc’ is misspelled twice (’specific experts’ and ’specific experts’). Please correct to ’specific’.
- **[writing]** In Section 3 (Introduction), the sentence ’By separating capability-specific training from cross-capability consolidation, OPD avoids the gradient conflicts caused by capability divergence’ ends with a stray comment marker ’%’ followed by a newline, breaking the flow. Remove the comment or integrate the text.
- **[writing]** In Section 5.1 (Main Results), the phrase ’surpassing the specific experts on both sides’ is slightly ambiguous. Consider clarifying to ’surpassing the individual domain-specific experts’ for precision.